

# Hwaran Lee

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| CONTACT INFORMATION           | NAVER AI Lab at NAVER Corp.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <a href="mailto:hwaran.lee@gmail.com">hwaran.lee@gmail.com</a> | <a href="https://github.com/hwaranlee">hwaranlee.github.io</a>          |
| RESEARCH INTERESTS            | Broad natural language-related topics including Language Models, Language Understanding and Generation, Dialog Systems, Fact Verification, Ethics for AI, Speech Recognition Systems, Neural Networks and Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                |                                                                         |
| EDUCATION                     | <b>KAIST</b><br>Ph.D., Electrical Engineering<br>Dissertation: <i>Neural Representations for Speech Recognition and Natural Language Generation</i><br>Advisor: Prof. Soo-Young Lee                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Daejeon, South Korea<br>Mar. 2013 – Aug. 2018                  |                                                                         |
|                               | <b>KAIST</b><br>B.S., Mathematical Science<br>Minor: Financial Engineering Program<br><i>Magna Cum Laude</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Daejeon, South Korea<br>Feb. 2008 – Aug. 2012                  |                                                                         |
| RESEARCH AND WORK EXPERIENCES | <b>Research Scientist</b> <ul style="list-style-type: none"><li>NAVER AI LAB, NAVER Corp., Seongnam, South Korea</li></ul> <b>Research Scientist</b> <ul style="list-style-type: none"><li>T-Brain, AI Center, SK Telecom, Seoul, South Korea</li></ul> <b>Graduate Research Assistant</b> <ul style="list-style-type: none"><li>Electrical Engineering, KAIST, Daejeon, South Korea</li></ul> <b>Undergraduate Researcher</b> <ul style="list-style-type: none"><li>Brain Science Research Center, Daejeon, South Korea</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mar. 2021 – Present                                            | Nov. 2018 – Feb. 2021<br>Mar. 2013 – Aug. 2018<br>Jun. 2012 – Feb. 2013 |
| PUBLICATIONS                  | <b>International Journal</b><br><br>[J1] <b>Hwaran Lee</b> , Seokhwan Jo, HyungJun Kim, Sangkeun Jung, and Tae-Yoon Kim, “SUMBT+LaRL: Effective Multi-domain End-to-end Neural Task-oriented Dialog System”, <i>IEEE Access</i> , 9 (2021): 116133-116146.<br><br>[J2] Geonmin Kim, <b>Hwaran Lee</b> , Bo-Kyeong Kim, Sang-Hoon Oh, and Soo-Young Lee, “Unpaired Speech Enhancement by Acoustic and Adversarial Supervision for Speech Recognition”, <i>IEEE Signal Processing Letters</i> , (2019): 159-163.<br><br>[J3] Ho-Gyeong Kim, <b>Hwaran Lee</b> , Geonmin Kim, Sang-Hoon Oh, and Soo-Young Lee, “Rescoring of N-best Hypotheses using Top-down Selective Attention for Automatic Speech Recognition”, <i>IEEE Signal Processing Letters</i> , (2018): 199-203.<br><br>[J4] <b>Hwaran Lee</b> , Geonmin Kim, Ho-Gyeong Kim, Sang-Hoon Oh, and Soo-Young Lee, “Deep CNNs Along the Time Axis With Intermap Pooling for Robustness to Spectral Variations”, <i>IEEE Signal Processing Letters</i> 23.10 (2016): 1310-1314.<br><br>[J5] <b>Hwaran Lee</b> , Nadeem Iqbal, Wonil Chang, and Soo-Young Lee, “A Calibration Method for Eye-Gaze Estimation Systems Based on 3D Geometrical Optics”, <i>IEEE Sensors Journal</i> 13, no. 9 (2013): 3219-3225. |                                                                |                                                                         |

- [J6] Nadeem Iqbal, **Hwaran Lee**, and Soo-Young Lee, “Smart User Interface for Mobile Consumer Devices Using Model-Based Eye-Gaze Estimation”, *IEEE Transactions on Consumer Electronics* 59, no. 1 (2013): 161-166.

#### International Conference

- [C1] John Yoon Young Chung, Wooseok Kim, Kang Min Yoo, **Hwaran Lee**, Eytan Adar, Minsuk Chang, “TaleBrush: Sketching Stories with Generative Pretrained Language Models”, In Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems, 2022 (to appear)
- [C2] Gi-Cheon Kang, Junseok Park, **Hwaran Lee**, Byoung-Tak Zhang, and Jin-Hwa Kim, “Reasoning Visual Dialog with Sparse Graph Learning and Knowledge Transfer”, Findings of the Association for Computational Linguistics: **EMNLP**, 2021.
- [C3] **Hwaran Lee**<sup>\*</sup>, Jinsik Lee<sup>\*</sup>, and Tae-Yoon Kim, “SUMBT: Slot-Utterance Matching for Universal and Scalable Belief Tracker”, *The 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.
- [C4] Geonmin Kim, **Hwaran Lee**, Bo-Kyeong Kim, and Soo-Young Lee, “Compositional Sentence Representation from Character within Large Context Text”, *International Conference on Neural Information Processing (ICONIP)*, 2017.
- [C5] Ho-Gyeong Kim, Jihyeon Roh, **Hwaran Lee**, Geonmin Kim, and Soo-Young Lee, “Active Learning for Large-scale Object Classification: from Exploration to Exploitation” *In Proceedings of the 3rd International Conference on Human-Agent Interaction, (HAI)*, 2015.
- [C6] Bo-Kyeong Kim, **Hwaran Lee**, Jihyeon Roh, and Soo-Young Lee, “Hierarchical committee of deep CNNs with exponentially-weighted decision fusion for static facial expression recognition”, *In Proceedings of the 2015 ACM on International Conference on Multimodal Interaction (ICMI)*, 2015.

#### International Workshop

- [W1] Geonmin Kim<sup>\*</sup>, **Hwaran Lee**<sup>\*</sup>, CheongAn Lee, Eunmi Hong, Byunggeun Kim, Soo-Young Lee, “A Deep Chatbot for QA and Chitchat.”, The Conversational Intelligence Challenge section on NIPS 2017 Competition Track Workshop, 2017.
- [W2] **Hwaran Lee**, Geonmin Kim, Jihyeon Roh, and Soo-Young Lee, “Learning Tonotopically Organized Auditory Feature-map from Speech by an Intermap Pooling Layer in a Deep CNN”, *15th China-Japan-Korea Joint Workshop on Neurobiology and Neuroinformatics (NBNI)*, 2015. (only abstract)
- [W3] Geonmin Kim, **Hwaran Lee**, Jaemyung Yu, and Soo-Young Lee, “Spoken Sentence Embedding from Character by Jointly Learning Character-level Compositional Word Model and RNN Sentence Encoder”, *15th China-Japan-Korea Joint Workshop on Neurobiology and Neuroinformatics (NBNI)*, 2015. (only abstract)

#### PATENTS

- [P1] Tae-Yoon Kim, Jin Kim, Hyungjoon Kim, Jinsik Lee, **Hwaran Lee**, Heewon Jeon, Seokhwan Jo, “Method and Apparatus for Providing Hybrid Intelligent Customer Consultation”, Korea Patent Application 10-2019-0136035, filed October 2019, Patent Pending.
- [P2] **Hwaran Lee**, Jinsik Lee, Tae-Yoon Kim, “Method and Apparatus for Dialogue State Tracking for Use in Goal-oriented Dialog System”,

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<sup>\*</sup>these authors contributed equally to this work

- Korea Patent Application 10-2019-0086380, filed July 2019, Patent Pending.
- PCT/KR2020/008832, filed July 2020, Patent Pending.

## RESEARCH PROJECTS

### **Meta Learner Project** 2020 - 2021

- Funded by SK Telecom
  - Researched and developed semi-supervised learning algorithms and modules for AutoTrainer
  - Tech leader for research and development of AutoML for various natural language processing tasks

### **End-to-end Goal-Oriented Dialog Systems** 2018 - 2019

- Funded by SK Telecom
  - Researched and developed a scalable multi-domain natural language understanding
  - Researched and developed end-to-end neural goal-oriented dialog systems
  - Participated in the End-to-End Multi-Domain Task-Completion Task (DSTC8 Track1 Task1 at AAAI 2020) and ranked 6th place

### **A Deep Chatbot for QA and Chitchat** 2017

- Project Leader, Funded by *Institute for Information & Communications Technology Promotion (IITP)*
  - Researched and developed an article-based chatbot that carry on both question-answering and chitchat conversations
  - **Ranked 3rd Place** in the Conversational Intelligence Challenge of NIPS 2017 Live Competition

### **Deep Learning Based Korean Understanding and Applications** 2015 – 2018

- Project Leader, Funded by HANCOM Inc.
  - Researched and developed Korean morpheme, syllable-level language models, sentence and document models
  - Researched and developed continual learning algorithms for language models

### **Speech Feature Extraction Based on Deep Learning for Continuous Speech Recognition Systems** 2013 – 2014

- Project Leader, Funded by LG Electronics
  - Developed convolutional neural networks for acoustic models of continuous English speech
  - Programmed C/C++ and CUDA based on KALDI toolkit

### **Development of Dialog-based Spontaneous Speech Interface Technology on Mobile Platforms** 2013 – 2014

- Funded by Electronics and Telecommunications Research Institute (*ETRI*)
  - Developed deep belief networks and auto-encoders for acoustic models of Korean syllable data
  - Programmed models in C/C++ and CUDA languages

## HONORS AND AWARDS

- Annual Roll Award, KAIST EE Apr. 2018
- Ranked 3rd, ConvAI challenge, NIPS 2017 Competition Track Workshop 2017
- Challenge Winner, ICMI EmotiW2015 2015
- Best Paper Award, HAI 2015
- Qualcomm Innovation Award 2015
- BK21 Plus Financial Support for Graduates Long Term Training May. 2014
- KAIST Graduate Scholarship Mar. 2013 - Aug. 2018

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|                     | <ul style="list-style-type: none"> <li>• Australian Endeavour Student Exchange Grant (AUD\$ 5000), Apr. 2011<br/>The University of Queensland</li> <li>• National Excellence Scholarship, KOSAF Feb.2008 - Feb. 2012</li> </ul>                                                                                                                                                                                                                                                                                  |
| ACADEMIC SERVICES   | <ul style="list-style-type: none"> <li>• Program Committee <ul style="list-style-type: none"> <li>• ARR 2021, ACL 2021, EMNLP 2021, COLING 2020</li> <li>• NeurIPS 2021, ICLR 2021-2022</li> <li>• WWW 2022</li> </ul> </li> <li>• Samsung Humantech Paper Awards Committee 2020</li> <li>• Qualcomm Innovation Awards Committee 2019</li> <li>• Speech Communication 2019</li> <li>• IEEE Transactions on Neural Networks and Learning Systems 2017 - 2018</li> <li>• Neural Processing Letters 2015</li> </ul> |
| INVITED TALKS       | <ul style="list-style-type: none"> <li>• Introduction to deep learning for dialogue systems <div style="text-align: right;"> Inha Univ., Seoul, Oct. 2020<br/> Yeonsei Univ., Seoul, Jun. 2020<br/> Sookmyung Women's Univ., Seoul, Oct. 2019 </div> </li> <li>• Toward end-to-end neural dialog systems for multi-domain task completion <div style="text-align: right;"> KAIST, Daejeon, Dec. 2019 </div> </li> </ul>                                                                                          |
| TEACHING EXPERIENCE | <p><b>Teaching Assistant</b></p> <ul style="list-style-type: none"> <li>• ChatBot PyTorch Implementation, KB-KAIST AI Sep. 2017</li> <li>• EE476 Audio-Visual Perception Models Spring 2016, 2017</li> <li>• EE538 Neural Networks Fall 2015, 2016, 2017</li> </ul> <p><b>Graduate Mentor</b></p> <ul style="list-style-type: none"> <li>• Undergraduate Mentoring Program, KAIST Counselling Center<br/>Fall 2014, Spring 2016</li> </ul>                                                                       |
| SKILLS              | <ul style="list-style-type: none"> <li>• <b>Languages</b> Python, Lua, MATLAB, C/C++, CUDA</li> <li>• <b>Libraries</b> PyTorch, TensorFlow, Torch7, KALDI</li> </ul>                                                                                                                                                                                                                                                                                                                                             |
| REFERENCES          | Prof. Soo-Young Lee (Ph.D. advisor)<br>Emeritus Professor, School of Electrical Engineering, KAIST sylee@kaist.ac.kr<br>Director, KAIST Institute for Artificial Intelligence <a href="http://cnsl.kaist.ac.kr">http://cnsl.kaist.ac.kr</a>                                                                                                                                                                                                                                                                      |